

ORF 309 Homework 4 - Question 5

Easier Approach

Variance of Love Triangles - Alternative Method

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Variable Definitions

- n = the number of vertices in the random graph
- p = the probability that any given edge exists (each edge is independent)
- N = the total number of love triangles on the site (what we want the variance of)
- I_{ijk} = the **indicator variable** for the triangle on vertices $\{i, j, k\}$

Understanding Indicator Variables

An **indicator variable** is a random variable that equals **1** if some event happens, and **0** if it doesn't. We use the letter I (for "indicator") to denote these. For every possible triple of vertices (i, j, k) , we define:

$$I_{ijk} = \begin{cases} 1 & \text{if vertices } i, j, k \text{ form a triangle (all 3 edges present)} \\ 0 & \text{otherwise} \end{cases}$$

The total number of love triangles N is then just the **sum of all these indicators**:

$$N = \sum_{1 \leq i < j < k \leq n} I_{ijk}$$

This is a standard technique: instead of reasoning about a complicated count directly, decompose it into a sum of simple 0/1 variables that are each easy to analyze.

Key properties of indicators used in this solution:

1. $E[I_{ijk}] = p^3$ — A triangle requires all 3 edges, each existing independently with probability p , so $P(\text{triangle exists}) = p \cdot p \cdot p = p^3$.
2. $I_{ijk}^2 = I_{ijk}$ — Since I is either 0 or 1, squaring it changes nothing ($0^2 = 0$, $1^2 = 1$). This simplifies the $E[N^2]$ computation.
3. $E[I_{ijk}I_{lmr}] = P(\text{both triangles exist})$ — The expectation of a product of indicators equals the probability that both events occur. Whether the two triangles share edges determines whether they are independent:

- **No shared edges** (0 or 1 common vertices): all 6 edges are distinct, so the events are independent and $E[I_{ijk}I_{lmr}] = p^6$.
- **One shared edge** (2 common vertices): only 5 distinct edges are needed, so $E[I_{ijk}I_{lmr}] = p^5$.

Easier Approach: Using $E[N^2]$

Instead of computing all the covariances, we can use the simpler formula:

$$\text{Var}(N) = E[N^2] - (E[N])^2$$

This approach is more direct because we just need to compute $E[N^2]$, which counts **pairs of triangles**.

Step 1: Recall $E[N]$

From the previous homework:

$$E[N] = \binom{n}{3} p^3$$

Therefore:

$$(E[N])^2 = \left(\binom{n}{3} p^3 \right)^2 = \binom{n}{3}^2 p^6$$

Step 2: Compute $E[N^2]$

Recall that $N = \sum_{1 \leq i < j < k \leq n} I_{ijk}$ where I_{ijk} are indicator variables for each triangle.

To square N , we use the fact that squaring a sum means multiplying every term by every other term. For a simple example with three terms: $(a + b + c)^2 = a^2 + b^2 + c^2 + ab + ac + ba + bc + ca + cb$. In general:

$$\left(\sum_{\alpha} I_{\alpha} \right)^2 = \sum_{\alpha} \sum_{\beta} I_{\alpha} \cdot I_{\beta}$$

where α and β each range over all $\binom{n}{3}$ possible triangles. We can split this double sum into two cases based on whether α and β refer to the **same** triangle or **different** triangles:

$$\sum_{\alpha} \sum_{\beta} I_{\alpha} I_{\beta} = \underbrace{\sum_{\alpha} I_{\alpha} \cdot I_{\alpha}}_{\text{same triangle } (\alpha=\beta)} + \underbrace{\sum_{\alpha \neq \beta} I_{\alpha} \cdot I_{\beta}}_{\text{different triangles } (\alpha \neq \beta)}$$

Writing this out with our subscript notation:

$$N^2 = \sum_{\{i,j,k\}} I_{ijk}^2 + \sum_{\{i,j,k\} \neq \{\ell,m,r\}} I_{ijk} \cdot I_{lmr}$$

Now take the expectation of both sides (expectation distributes over sums by linearity):

$$E[N^2] = \underbrace{\sum_{\{i,j,k\}} E[I_{ijk}^2]}_{\text{(A) diagonal terms}} + \underbrace{\sum_{\{i,j,k\} \neq \{\ell,m,r\}} E[I_{ijk} \cdot I_{lmr}]}_{\text{(B) cross terms}}$$

We compute each part separately below.

First term: $\sum E[I_{ijk}^2]$

Since I_{ijk} is an indicator, $I_{ijk}^2 = I_{ijk}$, so:

$$E[I_{ijk}^2] = E[I_{ijk}] = p^3$$

There are $\binom{n}{3}$ triangles, so:

$$\sum_{\{i,j,k\}} E[I_{ijk}^2] = \binom{n}{3} p^3$$

Second term: $\sum E[I_{ijk} I_{lmr}]$ for different triangles

We need $E[I_{ijk} I_{lmr}] = P(\text{both triangles exist})$.

This depends on how many vertices the two triangles share:

Case 1: Triangles share 0 or 1 vertices (no common edges)

If they have no edges in common, the events are independent:

$$E[I_{ijk} I_{lmr}] = P(I_{ijk} = 1) \cdot P(I_{lmr} = 1) = p^3 \cdot p^3 = p^6$$

How many such pairs?

Total pairs of different triangles: $\binom{\binom{n}{3}}{2} = \frac{1}{2} \binom{n}{3} (\binom{n}{3} - 1)$

We need to subtract the pairs that share 2 or 3 vertices (computed below).

Case 2: Triangles share exactly 2 vertices (share 1 edge)

If triangles $\{i, j, k\}$ and $\{i, j, \ell\}$ share edge (i, j) , then for both to exist we need 5 edges total (not 6):

- Edge (i, j) - shared
- Edges $(i, k), (j, k)$ - for first triangle
- Edges $(i, \ell), (j, \ell)$ - for second triangle

Therefore:

$$E[I_{ijk} I_{ij\ell}] = p^5$$

How many such pairs?

For each edge (i, j) :

- There are $\binom{n}{2}$ possible edges
- For each edge, we can form triangles by adding a third vertex
- There are $n - 2$ choices for the third vertex
- Number of pairs of triangles sharing this edge: $\binom{n-2}{2}$

Total pairs sharing an edge:

$$\binom{n}{2} \cdot \binom{n-2}{2} = \frac{n(n-1)}{2} \cdot \frac{(n-2)(n-3)}{2} = \frac{n(n-1)(n-2)(n-3)}{4}$$

Case 3: Triangles share 3 vertices (same triangle)

This is impossible for different triangles, already counted in the first term.

Step 3: Put it together

Number of pairs sharing 0 or 1 vertices:

$$\begin{aligned} &= (\text{Total pairs of triangles}) - (\text{Pairs sharing an edge}) \\ &= \binom{\binom{n}{3}}{2} - \binom{n}{2} \binom{n-2}{2} \end{aligned}$$

Therefore:

$$\begin{aligned} E[N^2] &= \binom{n}{3} p^3 + \left[\binom{\binom{n}{3}}{2} - \binom{n}{2} \binom{n-2}{2} \right] p^6 + \binom{n}{2} \binom{n-2}{2} p^5 \\ &= \binom{n}{3} p^3 + \binom{\binom{n}{3}}{2} p^6 - \binom{n}{2} \binom{n-2}{2} p^6 + \binom{n}{2} \binom{n-2}{2} p^5 \\ &= \binom{n}{3} p^3 + \binom{\binom{n}{3}}{2} p^6 + \binom{n}{2} \binom{n-2}{2} (p^5 - p^6) \\ &= \binom{n}{3} p^3 + \binom{\binom{n}{3}}{2} p^6 + \binom{n}{2} \binom{n-2}{2} p^5 (1 - p) \end{aligned}$$

Step 4: Compute the variance

$$\begin{aligned} \text{Var}(N) &= E[N^2] - (E[N])^2 \\ &= \binom{n}{3} p^3 + \binom{\binom{n}{3}}{2} p^6 + \binom{n}{2} \binom{n-2}{2} p^5 (1 - p) - \binom{n}{3}^2 p^6 \end{aligned}$$

Simplify using $\binom{\binom{n}{3}}{2} = \frac{1}{2} \binom{n}{3} (\binom{n}{3} - 1) = \frac{1}{2} \binom{n}{3}^2 - \frac{1}{2} \binom{n}{3}$:

$$\begin{aligned} \text{Var}(N) &= \binom{n}{3} p^3 + \left[\frac{1}{2} \binom{n}{3}^2 - \frac{1}{2} \binom{n}{3} \right] p^6 + \binom{n}{2} \binom{n-2}{2} p^5 (1 - p) - \binom{n}{3}^2 p^6 \\ &= \binom{n}{3} p^3 + \frac{1}{2} \binom{n}{3}^2 p^6 - \frac{1}{2} \binom{n}{3} p^6 - \binom{n}{3}^2 p^6 + \binom{n}{2} \binom{n-2}{2} p^5 (1 - p) \\ &= \binom{n}{3} p^3 - \frac{1}{2} \binom{n}{3} p^6 - \frac{1}{2} \binom{n}{3}^2 p^6 + \binom{n}{2} \binom{n-2}{2} p^5 (1 - p) \\ &= \binom{n}{3} p^3 \left(1 - \frac{1}{2} p^3 - \frac{1}{2} \binom{n}{3} p^3 \right) + \binom{n}{2} \binom{n-2}{2} p^5 (1 - p) \end{aligned}$$

Actually, let's simplify more cleanly:

$$\begin{aligned} \text{Var}(N) &= \binom{n}{3} p^3 - \frac{1}{2} \binom{n}{3} p^6 - \frac{1}{2} \binom{n}{3}^2 p^6 + \binom{n}{2} \binom{n-2}{2} p^5 (1 - p) \\ &= \binom{n}{3} p^3 (1 - p^3) - \frac{1}{2} \binom{n}{3}^2 p^6 + \binom{n}{3} p^3 \cdot p^3 + \binom{n}{2} \binom{n-2}{2} p^5 (1 - p) \\ &= \binom{n}{3} p^3 (1 - p^3) + \binom{n}{2} \binom{n-2}{2} p^5 (1 - p) \end{aligned}$$

Wait, let me recalculate more carefully. Actually, the cleanest form is:

$$\text{Var}(N) = \binom{n}{3} p^3 (1 - p^3) + \binom{n}{2} \binom{n-2}{2} p^5 (1 - p)$$

This matches our previous answer!

Why this approach is easier

1. **More direct:** We compute $E[N^2]$ by counting pairs of triangles, rather than tracking individual covariances
2. **Cleaner bookkeeping:** Only need to classify pairs by overlap (0-1 vertices vs 2 vertices)
3. **Single formula:** Everything flows from $\text{Var}(N) = E[N^2] - (E[N])^2$
4. **Same result:** Both methods give the same answer, but this one feels less tedious

Key insight

When dealing with sums of indicators:

- Computing $E[N^2]$ means counting **ordered pairs of outcomes**
- For indicators, $E[I_i I_j] = P(\text{both occur})$
- The trick is to **count by cases** based on overlap structure

This is essentially the same as the covariance approach, but with less explicit covariance notation!